

Luminal Transmission of Decapentaplegic in *Drosophila* Imaginal Discs

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Summary

Drosophila imaginal discs are sac-like appendage primordia comprising apposed peripodial and columnar cell layers. Cell survival in disc columnar epithelia requires the secreted signal Decapentaplegic (DPP), which also acts as a gradient morphogen during pattern formation. The distribution mechanism by which secreted DPP mediates global cell survival and graded patterning is poorly understood. Here we report detection of DPP in the luminal cavity between apposed peripodial and columnar cell layers of both wing and eye discs. We show that peripodial cell survival hinges upon DPP signal reception and implicate DPP-dependent viability of the peripodial epithelium in growth of the entire disc. These results are consistent with luminal transmission of the DPP survival signal during imaginal disc development.

Introduction

Organ and limb size are defined by the complex interplay between pattern formation and proliferative growth. Coordination of these processes necessitates signaling among cells within an epithelium as well as signaling between adjacent cell layers to ensure proportionate size of organ structures derived from distinct cell sheets. A prime example of transepithelial size coordination is the developing amphibian eye, where apposed optic cup and lens primordia regulate their size in proportion to one another (Harrison, 1929; Twitty and Elliot, 1934; Rothman and Spemann, 1938). Similarly, multiple inductive interactions between presumptive mesoderm and apposed ectoderm coordinate growth in the vertebrate limb bud (Harrison, 1935; Niswander et al., 1993; Fallon et al., 1994; Tickle, 1995).

The eyes and appendages of *Drosophila* develop from epidermal invaginations called imaginal discs. During larval development, discs are two-sided sacs comprising a columnar cell layer and an overlying squamous peripodial epithelium (Auerbach, 1936; Cho et al., 2000; Gibson and Schubiger, 2000). During larval development, apposed peripodial and columnar epithelia grow in proportion, suggesting coordinate regulation by a singular size control mechanism. In this light it is interesting to consider the diminutive discs of *decapentaplegic* (*dpp*) mutant flies. On the basis of published images

(e.g., Spencer et al., 1982; Bryant, 1988) and our own observations, it seems that loss of *dpp* results in proportionate size reduction in both imaginal disc cell layers. Supporting this view, Bryant (1988) reports excessive cell death in both peripodial and columnar epithelia of *dpp* mutant discs. Further, Cho et al. (2000) report that *dpp* is predominantly expressed in peripodial cells during eye disc development. These observations are consistent with a function for DPP in the growth of both peripodial and columnar cell layers.

The *dpp* locus is named for a dramatic mutant phenotype, wherein development of 15 of the imaginal discs is severely compromised (Spencer et al., 1982). Molecular studies reveal that *dpp* encodes a member of the TGF- β family of extracellular signals (Padgett et al., 1987; Panganiban et al., 1990b) that acts at both short and long range to activate the transcription of target genes in receiving cells (Nellen et al. 1996; Lecuit et al., 1996). Detailed biochemical analyses show that DPP is received by a TGF- β type II transmembrane receptor, which heterodimerizes with the type I receptor Thickveins (TKV) upon ligand binding (Brummel et al., 1994; Nellen et al., 1994; Penton et al., 1994; Wrana et al., 1994). DPP-induced activation of TKV leads to phosphorylation of the cytosolic transducer MAD, which then translocates to the nucleus either as a homodimer or as a heterodimer with Medea to regulate target gene transcription (Inoue et al., 1998; Das et al., 1998; Maduzia and Padgett, 1997). Recent studies indicate that DPP activates target loci by inhibiting Brinker-mediated transcriptional repression in the wing blade primordium (Campbell and Tomlinson, 1999; Minami et al., 1999; Marty et al., 2000; Affolter et al., 2001).

In the wing disc, DPP is transcribed in a thin stripe along the anterior-posterior compartment boundary (Posakony et al., 1990). Diffusing from these source cells, secreted DPP is proposed to form a planar morphogen gradient across the A/P axis of the disc columnar epithelium (Zecca et al., 1995; Lawrence and Struhl, 1996; Entchev et al., 2000; Teleman and Cohen, 2000). Supporting the role of DPP as a true morphogen, DPP signaling is shown to activate transcriptional targets at distinct threshold concentrations (Lecuit et al., 1996; Nellen et al., 1996). However, despite graded activity in pattern formation, DPP is broadly required for cell survival/proliferation throughout the wing blade and eye primordia (Burke and Basler, 1996a, 1996b). Recent studies indicate that columnar cells impaired for DPP reception or transduction are eliminated by a *brinker*-dependent process of cell competition (Moreno et al., 2002). The dual mechanism by which a planar morphogen gradient of DPP might govern ubiquitous survival and concentration-dependent patterning has remained elusive, although recent evidence suggests that disc cell proliferation is not dependent on the ability of cells to detect a DPP gradient (Martin-Castellanos and Edgar, 2002). Embryonic functions of DPP could provide some insight into its potential mode of action in discs. DPP mediates communication between apposed cell layers during endoderm induction (Panganiban et al., 1990a;

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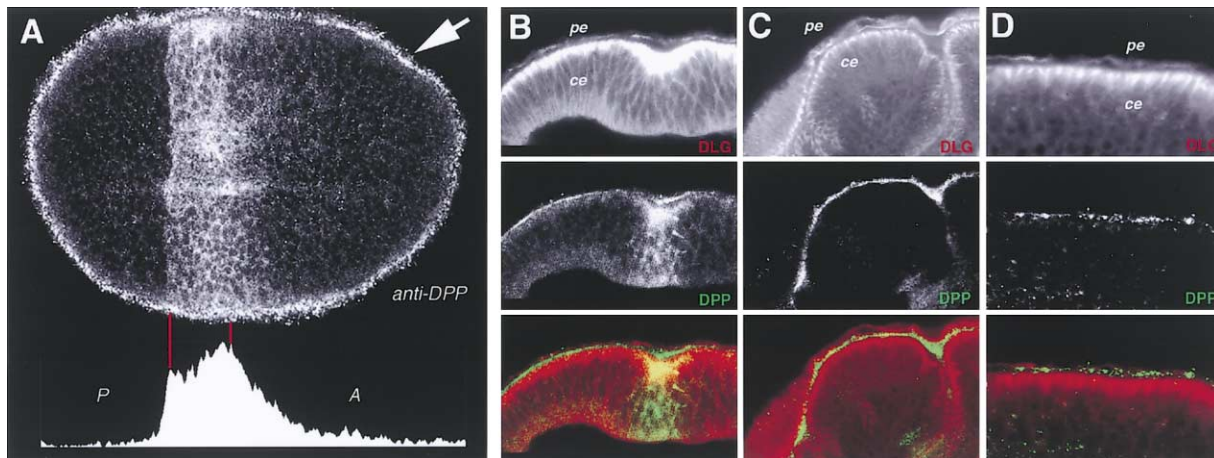


Figure 1. Immunolocalization of DPP in Third Instar Imaginal Discs

(A) Distribution of DPP in an optical section through columnar cells of the blade primordium of a third instar wing disc. DPP is detected in a stripe along the disc A/P boundary and in a compartmentally asymmetric gradient pattern. Below, a pixel intensity plot shows anti-DPP in an asymmetric gradient along the A/P axis. A ring of anti-DPP staining (arrow) is observed where the optical section intersects with the luminal space apical to the columnar cell layer.

(B–D) Extracellular DPP is detected in the luminal cavity between peripodial (pe) and columnar epithelia (ce). All images are optical cross-sections through material stained for DLG (red) to outline cell boundaries and DPP (green).

(B) In the eye, expression of DPP is apparent in columnar cells of the morphogenetic furrow and throughout the luminal cavity.

(C) In an optical cross-section through a second leg disc, DPP staining is clearly luminal above the end knob region, primordia for the tarsal segments of the leg.

(D) DPP in the restricted luminal space between peripodial and columnar epithelia in the center of the wing blade primordium. Discs were mounted on their lateral edges to obtain the optical cross-sections in (C) and (D).

Immergluck et al., 1990; Bienz, 1997). Also during embryogenesis, dorsal ectodermal cells secrete DPP to induce cardiac fate in underlying mesoderm (Frasch, 1995). In both of these cases, DPP is employed to transmit positional information outside the context of a monolayer epithelial sheet.

Here we demonstrate that DPP is present in the luminal cavity separating apposed peripodial and columnar cell layers of eye, leg, and wing imaginal discs. We also show that removal of the TKV receptor from peripodial cells renders them inviable and morphologically abnormal in both eye and wing discs. In addition, peripodial-specific inhibition of DPP signaling causes size reduction and patterning defects in wing and eye discs. We suggest that luminal DPP signals to peripodial cells and that *dpp*-dependent growth of the peripodial epithelium is a necessary component of global disc morphogenesis. On the basis of these findings, we propose that spatial restriction of DPP in distinct luminal and epithelial domains could account for its dual role in growth and pattern formation.

Results

Luminal DPP in Wing, Leg, and Eye Discs

In the wing disc, DPP transcription is restricted to a strong band that runs through the center of the columnar epithelium and continues as a faint thin stripe along the extreme anterior edge of the peripodial cell layer (Posakony et al., 1990; data not shown). Despite this localized expression, a broad functional requirement for TKV demonstrates that imaginal disc cells receive DPP over a large range (Burke and Basler, 1996a; Martin-Castellanos and Edgar, 2002). Current models postulate

that DPP is distributed by an active process of planar transcytosis (Entchev et al., 2000). Conceptually, however, there are at least two alternate routes of DPP movement to all cells of the imaginal disc: “laterally” through epithelia or “vertically” through the disc lumen. We stained discs with an antibody directed to the DPP protein (Panganiban et al., 1990b) to distinguish between these two possibilities.

As expected, punctate intracellular DPP expression was observed in the known *dpp* expression domains of leg, wing, and eye disc columnar epithelia. For example, DPP was observed in a stripe and a compartmentally asymmetric distribution similar to the DPP pathway activity gradient in the wing disc (Figure 1A; Tanimoto et al., 2000; Teleman and Cohen, 2000). In the eye disc, strong DPP expression was observed in columnar cells of the morphogenetic furrow (Figure 1B), as previously reported (Heberlein et al., 1993). In addition, however, we observed intense extracellular DPP in the luminal cavities of eye, leg, and wing discs (Figures 1B–1D). Luminal DPP was detected in globular aggregates that did not form a visibly graded distribution, but, rather, appeared evenly spread throughout the luminal cavity. Given the unexpected nature of the results, we ruled out the possibility of a secondary antibody artifact using appropriate controls. To test the specificity of the primary antibody, we eliminated *dpp* expression in living discs using a temperature-sensitive mutation in *hedgehog* (*hh^{ts2}*; Ma et al., 1996). At the *hh*-permissive temperature, anti-DPP staining was broadly distributed (Figure 2A) and primarily luminal (Figure 2C). Under *hh*-restrictive conditions, anti-DPP staining was weak (Figure 2B) and luminal DPP was no longer visible (Figure 2D). These experiments support the specificity of anti-DPP

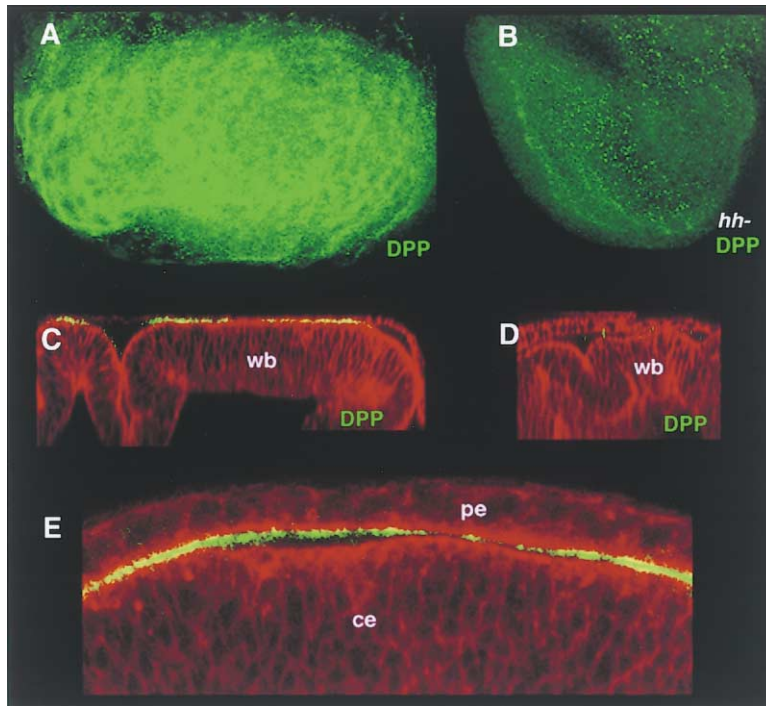


Figure 2. Luminal Anti-DPP Staining, in Green, Is *hh* Dependent in the Wing Imaginal Disc

Discs in (C), (D), and (E) were counterstained with rhodamine-phalloidin (red) to show cell outlines.

(A) Late third instar *hh^{ts2}* disc raised at a permissive temperature (18°C). The image is the accumulated projection of a confocal Z-series through the peripodial epithelium, lumen, and columnar cell layer of the wing blade primordium. Anti-DPP staining is uniform along the A/P axis.

(B) *hh^{ts2}* disc from a late third instar larva shifted to the nonpermissive temperature (29°C) 48 hr prior to dissection (*hh*[−]). Anti-DPP staining is mostly eliminated. The image is the accumulated projection of a confocal Z-series collected and processed under identical conditions to the control shown in (A).

(C) Confocal XZ section through a control *hh^{ts2}* disc raised at permissive temperature, revealing that anti-DPP staining is primarily luminal—an intense band along the apical surface of the columnar epithelium of the presumptive wing blade (wb). DPP levels within the columnar epithelium are relatively low compared with luminal DPP and not observed under these conditions.

(D) Confocal XZ section through a late third instar stage *hh^{ts2}* disc shifted to the nonpermissive temperature 48 hr prior to dissection. All anti-DPP staining is eliminated, including the intense luminal signal. The image was collected and processed under identical conditions to the control shown in (C).

(E) Anti-DPP staining is extracellular in an optical section through peripodial (pe) and columnar (ce) epithelia in the ventral region of the presumptive wing blade.

staining and show that DPP is broadly distributed and highly concentrated in the luminal space. Analysis of control wing discs counterstained with phalloidin demonstrates that luminal anti-DPP staining is indeed extracellular (Figure 2E). Previous studies also document the specificity of the antibody, which has been used to immunoprecipitate DPP from cell culture media (Panganiban et al., 1990b) and to detect extracellular DPP during signaling between apposed cell layers in the embryo (Panganiban et al., 1990a). However, it remains possible that this antibody recognizes the DPP prodomain and not the mature DPP signaling ligand. We therefore used a biologically active DPP-GFP fusion protein (Entchev et al., 2000) to analyze the spatial distribution of the mature DPP ligand.

Luminal Distribution of a DPP-GFP Fusion Protein

The authors of two recent papers have made use of DPP-GFP fusion proteins to address the extracellular distribution of the DPP ligand in wing discs (Teleman and Cohen, 2000; Entchev et al., 2000). Intriguingly, a fusion construct bearing the DPP cleavage and secretory transport sequences fused to GFP (sGFP) is secreted into the disc lumen upon expression in the endogenous DPP domain (Entchev et al., 2000). This indicates that the DPP cleavage and secretory transport sequences are sufficient to direct a naïve GFP reporter protein into the lumen. To confirm our observation of luminal DPP, we examined the distribution of a GFP-tagged version of the mature DPP signaling ligand in live wing discs (DPP-GFP; Entchev et al., 2000). Under the control of a disc-specific *dpp-Gal4* driver, DPP-GFP

was secreted into the luminal cavity (Figures 3A–3D). In confocal XZ sections, luminal fluorescence was considerably more intense than DPP-GFP observed within columnar cells immediately flanking the *dpp* expression domain (Figures 3C and 3D). Within cells of the columnar epithelium, intracellular DPP-GFP was observed in punctate apical structures (Figures 3B–3D), consistent with the report of Entchev et al. (2000). These apical structures were observed in close proximity to the lumen and could represent endocytic vesicles containing internalized luminal DPP-GFP. Since *dpp-Gal4* is expressed in a lateral stripe of peripodial cells in addition to columnar cells, the luminal DPP-GFP we observed could have been secreted from either or both disc cell layers. However, DPP-GFP also accumulated in the lumen when expressed from presumptive wing margin columnar cells under the control of *c96-Gal4* (data not shown; *c96-Gal4*, Gustafson and Boulianne, 1996).

TKV Is Required for Peripodial Cell Survival

Given the high levels of luminal DPP and the broad requirement for DPP signal transduction, we wondered whether DPP signaling was active in peripodial cells. To test this idea, we used the flp-FRT method of Xu and Rubin (1993) to generate peripodial cell clones lacking the DPP receptor TKV, which is absolutely required for DPP signaling in vivo (Ruberte et al., 1995). Peripodial *tkv⁵* null clones marked by the absence of GFP were induced during larval development. On the basis of twin spot analysis, 77% of peripodial *tkv⁵* null clones induced at 48 hr after egg laying (AEL) did not survive to the late third instar (*n* = 48 clones; Figures 3E and 3F). Similarly,

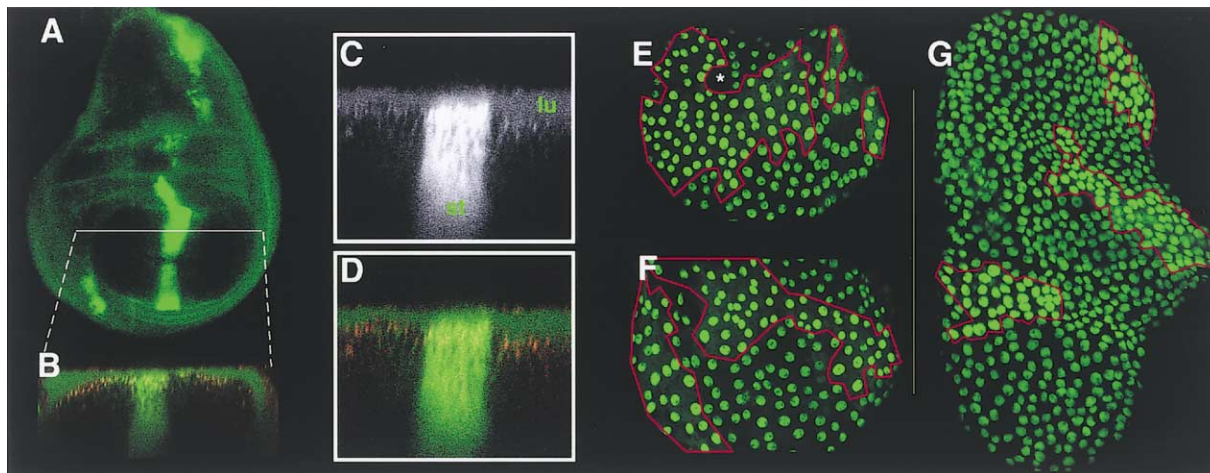


Figure 3. DPP-GFP Accumulates in the Luminal Space, and Peripodial Cell Survival Requires TKV

(A–D) DPP-GFP (green) is expressed under the control of *dpp-Gal4* in live third instar wing discs. Spacers were employed to prevent the coverslip from distorting the native tissue architecture.

(A) High levels of DPP-GFP are detected in the endogenous *dpp* expression domain of columnar cells.

(B) In XZ sections through the wing blade primordium at the position indicated by the solid white line in (A), strong fluorescence is detected throughout the luminal cavity. Discs were vitally counterstained with phalloidin (red), which weakly labeled cell outlines but intensely labeled punctate structures on the apical side of the disc columnar epithelium. Intriguingly, within columnar cells, DPP-GFP colocalized with the internalized phalloidin (yellow spots), consistent with the observation of DPP-GFP in apical punctate structures by Entchev et al. (2000).

(C) Higher magnification XZ section showing DPP-GFP distribution in the central wing blade primordium. The stripe (st) and luminal (lu) fractions of DPP-GFP are indicated. Luminal DPP-GFP intensity is significantly higher than in columnar cells immediately adjacent to the *dpp* stripe domain.

(D) Same disc as in (C), with vital phalloidin stain (red) to label apical punctate structures, which appear as red dots along the luminal surface of the columnar epithelium.

(E–G) *tkv*⁵ mutant cell clones marked by loss of Ubiquitin-GFP (green) are not viable in peripodial epithelia apposed to the wing blade (E and F) or eye (G) primordia. Shown here are live discs mounted in *Drosophila* Ringer's solution to better maintain the structure of the peripodial epithelium for imaging in a single plane. Wild-type 2 × GFP twin spots are outlined in red. GFP-negative, *tkv*⁵/*tkv*⁵ twin spots have been eliminated from the peripodial cell layer. A possible diminutive *tkv* mutant clone is indicated by the asterisk in (E).

78% of peripodial *tkv*⁵ mutant clones induced at 55 hr AEL were eliminated ($n = 27$). We verified the peripodial requirement for *tkv* with a different mutant allele (*tkv*⁷). Of 15, 2 × GFP twin spots in peripodial cells over the wing blade primordium, only 2 were associated with a *tkv*⁷ clone of comparable size. We conclude that DPP signaling is required for survival of peripodial cells apposed to the presumptive wing blade during larval development. Impairment of DPP signaling most likely triggers peripodial clone elimination by a cell competition mechanism similar to that observed in columnar cells (Moreno et al., 2002). Intriguingly, we did observe frequent *tkv* mutant peripodial cell clones over the presumptive hinge and notum, suggesting that there could be regional differences in the requirement for peripodial DPP signaling. Similar regional requirements for *tkv* are observed in wing disc columnar cells (Burke and Basler, 1996a).

To explore the general requirements for peripodial DPP signal transduction, we also analyzed the requirement for *tkv* in developing eye discs. Here it should be noted that there is a dynamic pattern of *dpp* expression in eye peripodial cells throughout larval development (Cho et al., 2000), and peripodial requirements for *tkv* could reflect either *cis*- or *trans*-epithelial signaling. Still, consistent with a general requirement for DPP signaling in peripodial cells, 67% of *tkv*⁵ mutant cell clones were eliminated from the eye peripodial epithelium (Figure 3G; $n = 45$ clones induced at 60 AEL).

Aberrant Size and Morphology of *tkv*⁵/*tkv*⁵

Peripodial Cells

In the clonal analyses presented above, a considerable fraction of *tkv* mutant peripodial clones survived to the late third instar. However, most surviving *tkv* mutant clones exhibited abnormal cell size, clone size, and clone morphology (Figures 4A–4D), documenting a cell-autonomous requirement for *tkv*, even in the absence of elimination. Some viable *tkv* clones in the eye peripodial epithelium appeared to be morphologically normal (Figure 4C), but the position of these clones was always biased to regions of the dorsal eye and interantennal connective ($n = 10$). This phenomenon could reflect a stronger requirement for TKV in ventral eye peripodial cells. The result is interesting in consideration of viable *dpp* mutants in which the ventral eye is more strongly affected (Spencer et al., 1982; Chanut and Heberlein, 1997).

Engineering a Wing Peripodial Gal4 Driver

Our observations thus far are consistent with a role for luminal DPP in promoting peripodial cell survival. To functionally dissect the role of peripodial DPP signal transduction in overall disc morphogenesis, we used the Gal4/UAS system (Brand and Perrimon, 1993) to direct sustained antagonism of DPP signaling in wing peripodial cells. This was accomplished by misexpressing a physiological inhibitor of DPP signaling, *daughters against DPP* (*dad*; Tsuneizumi et al., 1997), which is

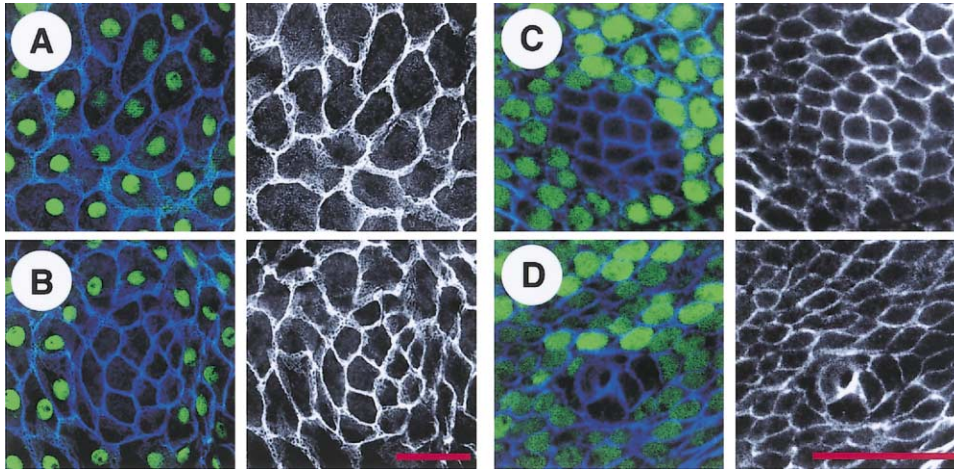


Figure 4. *tkv*⁵ Mutant Cell Clones Display Reduced Cell Size and Aberrant Shape

Color images at left show Discs Large (DLG; blue) to mark cell outlines (after Cho et al., 2000) and Ub-GFP (green) to mark nonmutant cells. Images at the right show DLG alone in black and white.

(A) Wild-type GFP-expressing peripodial cells above the presumptive wing blade.

(B) A GFP-negative *tkv*⁵ mutant clone is visible in a field of wild-type peripodial cells directly above the wing blade primordium. The *tkv* mutant cells are considerably smaller than surrounding cells. In addition, the clone is rounded up, perhaps reflecting adhesive differences with neighboring cells.

(C) *tkv*⁵ clones are more frequently viable on the dorsal side of the eye peripodial epithelium. Eye peripodial cell clones tend to grow in lines parallel to the disc D/V axis; this dorsal *tkv*⁵ mutant cell clone is rounded up, but the cells are otherwise indistinguishable from neighbors.

(D) Aberrant *tkv*⁵ clone size and morphology in the ventral peripodial epithelium of the same eye disc shown in (C).

endogenously expressed in a central domain of the wing columnar epithelium. We generated a Gal4 driver with peripodial-specific expression in the wing disc by converting a wing peripodial-specific *P{LacZ}* insertion strain (*P1615*; Russell et al., 1998) to a *Gal4* strain by homologous recombination between injected *P{w⁺, Gal4}* and genomic *P{LacZ}* sequences during germline transformation of *w*; *P1615* embryos. One stable transformant, *AGiR-Gal4*, drove expression of *UAS-GFP* throughout the wing peripodial epithelium and not in columnar cells of the presumptive wing blade (Figures 5A–5D). Some limited GFP expression was also detected in patches of cells in the presumptive hinge and notum. Confirming that targeted conversion was successful, chromosomal *in situ* hybridization and plasmid rescue localized the *AGiR-Gal4* insertion to 66D (data not shown), the cytological position of *P1615* (The Fly-base Consortium, 1999).

Peripodial DPP Signal Transduction Is Required for Wing Development

We inhibited DPP signal transduction in wing peripodial cells via overexpression of *UAS-dad* (Tsuneizumi et al., 1997). Experimental larvae displayed small and strikingly abnormal wing discs, often with deep ectopic clefts through the blade primordium (Figures 5E and 5F). Despite severe abnormalities, the peripodial epithelium remained structurally intact in these experiments, evidenced by the ability of *AGiR-Gal4>UAS-dad* discs to evaginate normally and give rise to adult cuticle. This observation indicates that the observed phenotypes were not the result of simply ablating the wing peripodial cell layer. *AGiR-Gal4>UAS-dad* adult wing phenotypes were consistent with the disc defects, including small

wings, small wings with duplicated margins, and, less frequently, small wings split in half along the A/P boundary (Figures 5G–5J). Wing blade size reduction was the most consistent phenotype. In *AGiR-Gal4>UAS-dad* animals shifted to 29°C from 48–72 hr AEL to increase activity of Gal4, 15/15 wings were narrower and shorter than wild-type controls. Along with size reduction, patterning defects in at least one wing were 44% penetrant in animals raised at 25°C (*n* = 34 pharate adults). This frequency increased to 100% in larvae shifted to 29°C for 24 hr during the early third instar (*n* = 20 pharate adults).

Intriguingly, *AGiR-Gal4>UAS-dad* wing pattern defects consistently localized to the D/V (Figure 5I) and A/P boundaries (Figure 5J). Ectopic cleft formation was always parallel to (but not necessarily coincident with) the A/P boundary in the columnar epithelium, and most split discs exhibited compartmentally asymmetric frequencies of cell division (Figure 6A). The developmental basis for these compartment-specific effects is not clear. We also used the distribution of Wingless (WG) to assess the integrity of the D/V boundary in *AGiR-Gal4>UAS-dad* wing discs and observed a wide range of aberrant WG staining patterns (Figures 6B and 6C). The diverse phenotypes observed in these experiments are probably best explained by excessive cell death within the disc columnar epithelium. This was confirmed by staining *AGiR-Gal4>UAS-dad* wing discs with the apoptosis indicator acridine orange (Figures 6D and 6E). Columnar cell death could be caused by loss of a DPP-dependent peripodial signal or, alternatively, constriction and aberrant morphogenesis of the wing blade primordium due to a failure in the coordinate expansion of the peripodial sac.

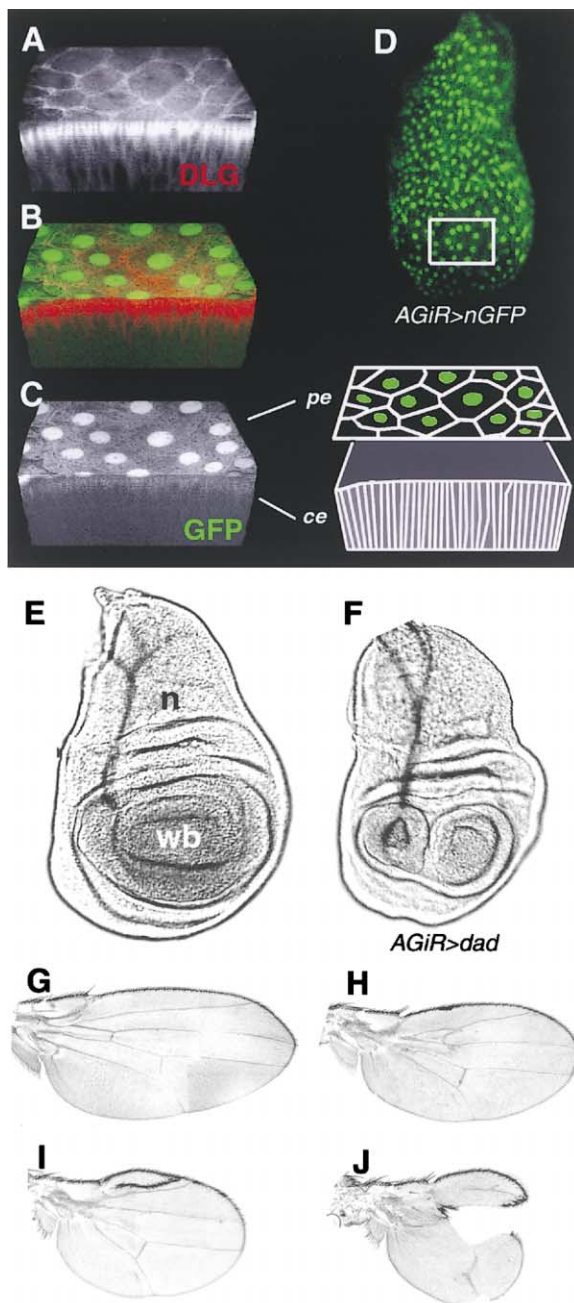


Figure 5. Peripodial DPP Signal Transduction Is Necessary for Wing Disc Growth and Patterning

(A–C) Peripodial specificity of *AGIR-Gal4* is demonstrated by computer-enhanced cross-sections through the presumptive wing blade of an *AGIR-Gal4 > UAS-GFP* larva.

(A) DLG ([B], red) marks cell outlines of both peripodial and columnar cells. Squamous peripodial cells are visible directly above the columnar epithelium.

(B) Merge of (A) and (C).

(C) Nuclear GFP is strictly peripodial in the GFP channel alone ([B], green). Cartoon illustrates the 3D tissue architecture and distinguishes peripodial (pe) and columnar (ce) epithelia.

(D) *AGIR-Gal4* drives *UAS-GFP* transgene expression throughout the wing disc peripodial epithelium, including peripodial cells over the presumptive wing blade and notum. A white box indicates the approximate area of detail depicted in (A)–(C).

(E–J) DPP-dependent growth of peripodial epithelia is required for wing morphogenesis.

Peripodial DPP Signal Transduction Is Required for Eye Development

To test the role of peripodial DPP signaling in the eye, we used an eye peripodial Gal4 driver to direct expression of *UAS-dad* (*c311-Gal4*; Gibson and Schubiger, 2000). Experimental animals exhibited reduced eye discs devoid of squamous peripodial cells and lacking any evidence of pattern formation or neuronal differentiation (Figures 7A and 7B). We analyzed the subcellular distribution of DLG (Discs Large; Woods and Bryant, 1991) to assess epithelial architecture. Normally, DLG is evenly distributed throughout the lateral cell membrane of peripodial cells, but preferentially localized to septate junctions on the apical side of the columnar cell layer (Woods et al., 1997; Figure 7C). Computer-enhanced cross-sections suggest that many *c311-Gal4 > UAS-dad* eye discs were comprised entirely of polarized epithelium (Figure 7D). It is unclear whether peripodial cells converted to columnar/cuboidal morphology or were simply eliminated in these experiments, but the results clearly suggest a requirement for peripodial DPP signaling in growth and epithelial morphogenesis in the eye imaginal disc.

Discussion

The Luminal Transmission of DPP

Observations from three classes of experiments suggest that DPP is secreted into the luminal cavity of developing imaginal discs and that DPP-dependent survival of the peripodial epithelium is required for eye and wing development. First, immunocytochemistry and direct observation of a GFP-tagged DPP molecule establish the luminal localization of DPP (Figures 1–3). Second, FLP/FRT clonal analyses demonstrate that eye and wing peripodial cells receive and require DPP signaling via *tkv* during disc development (Figures 3 and 4). Third, Gal4/UAS-mediated gene misexpression shows that peripodial-specific inhibition of *dpp* signaling results in severely reduced disc size (Figures 5–7). The limitations of each approach are several, but a conservative assessment of the data indicates that DPP is secreted into the lumen in close proximity to peripodial cells whose survival hinges on DPP reception. Peripodial transduction of DPP is in turn required for cell survival and coordinate growth of the peripodial sac during disc development. Inhibiting DPP signaling in the peripodial

(E) Control wing disc. The presumptive notum (n) and wing blade (wb) regions are indicated. Note that peripodial cells lying above the thickened disc columnar epithelium are essentially transparent and, thus, often overlooked.

(F) *AGIR-Gal4 > UAS-dad* wing disc exhibiting reduced size of the blade region and a deep ectopic cleft parallel to the A/P compartmental boundary.

(G) Wild-type wing blade from a pharate adult.

(H–J) Wing blades from *AGIR-Gal4 > UAS-dad* pharate adults are small and display severe patterning defects. Experimental animals were shifted to 29°C at 48 hr after egg deposition to increase Gal4 expression. Wing patterning phenotypes were surprisingly variable, but almost all show reduced growth, as in (H). In (I), the triple row is duplicated. In (J), the wing blade is bifurcated along the A/P boundary, corresponding to the A/P split discs of the type shown in (F).

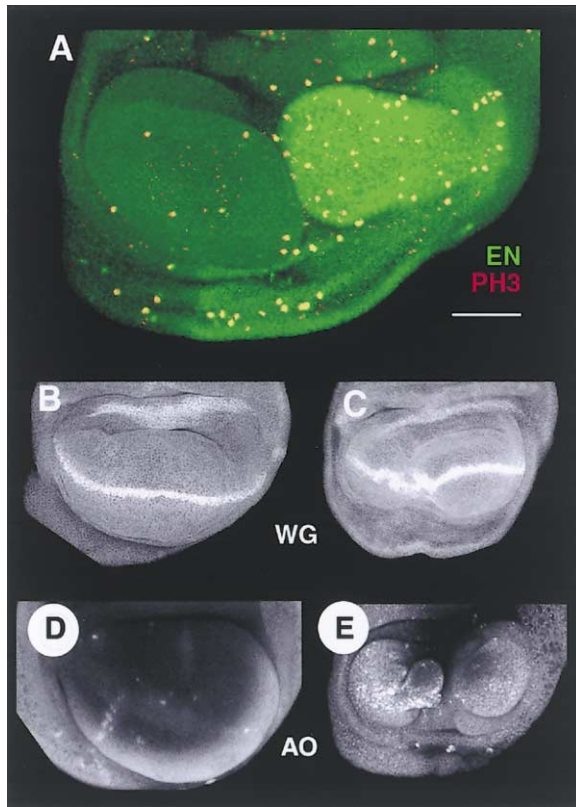


Figure 6. Peripodial Transduction of DPP Is Necessary for Pattern Formation and Cell Survival in the Wing Imaginal Disc

In the discs shown here, *AGIR-Gal4>UAS-dad* larvae were shifted to 29°C at 48 hr after egg deposition to increase Gal4 expression. (A) A deep cleft through the wing blade primordium runs parallel to the A/P compartment boundary in an *AGIR-Gal4>UAS-dad* wing disc stained with antibodies against EN and phosphohistone H3 to detect mitotic figures. Interestingly, the frequency of mitotic figures is asymmetric with respect to the A/P compartment boundary. (B) Control wing disc stained with an antibody against the WG protein. (C) A wide range of aberrant WG staining patterns are observed in *AGIR-Gal4>UAS-dad* wing blade primordia. The phenotype could result from violation of the D/V lineage restriction or, more likely, columnar cell death. (D) Control wing disc stained with acridine orange (AO) to detect cell death, which is minimal during normal development. (E) Excessive cell death is observed in the blade primordia of *AGIR-Gal4>UAS-dad* wing discs stained with acridine orange.

epithelium causes severe developmental defects, perhaps due to loss of a *dpp*-dependent secondary signal or morphometric constraints imposed by asynchronous growth of apposed peripodial and columnar cell layers.

While it remains a formal possibility that luminal DPP is biologically inert, several arguments favor the functionality of luminal DPP in signaling to peripodial cells and possibly to columnar cells, as well. First, DPP acts as a secreted signal between cell layers at several other times in *Drosophila* development; hence, the ability of functional DPP to move in the extracellular space between apposed cell layers is well established (e.g., Frasch, 1995). Second, active transport models poorly account for the generalized role of DPP in imaginal disc growth. The regulation of wing growth by a DPP morphogen gradient emanating from a geometrically central

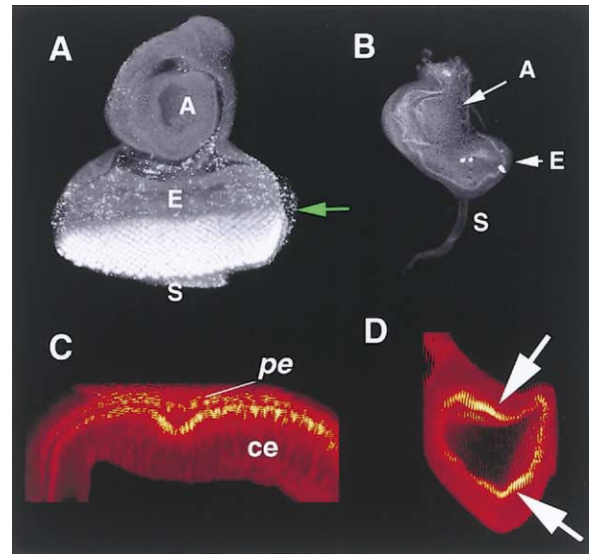


Figure 7. DPP-Dependent Growth of Peripodial Epithelia Is Required for Eye Morphogenesis

(A) Control eye-antennal disc stained for ELAV to detect photoreceptor recruitment behind the morphogenetic furrow (green arrow). Eye (E), antenna (A), and stalk (S) regions are easily distinguished. (B) Severely reduced growth is observed in *c311-Gal4>UAS-dad* eye-antennal discs. This disc was oriented on the basis of the position of the stalk. Only two or three photoreceptors express ELAV, indicating a defect in pattern formation. (C) Cross-sectional perspective of an eye imaginal disc stained for DLG (red). The disc is made up of distinct peripodial (pe) and columnar (ce) cell layers. DLG distribution is polarized in columnar epithelia but uniformly stains lateral cell boundaries in the peripodial cell layer. (D) Cross-sectional perspective of a *c311-Gal4>UAS-dad* eye disc. Global architecture of the disc is disrupted, with DLG revealing columnar-like morphology throughout the epithelium (arrows). The luminal cavity is enlarged, and the disc assumes a spheroid morphology. The squamous peripodial cells have either been eliminated or converted to a cuboidal/columnar cell shape.

stripe has great appeal. However, DPP signaling is also required for proliferative growth in the developing eye, where DPP is primarily expressed in peripodial cells during much of larval development (Cho et al., 2000). In the eye, DPP is not expressed in a central stripe but still maintains a key role in disc cell proliferation. We suggest that luminal transmission of DPP (versus planar active transport) can better account for the general role of DPP in growth of discs of different size and shape. Third, we report here the direct observation of luminal DPP and the requirement for DPP signaling via TKV in peripodial epithelia. Given the high levels of luminal DPP directly juxtaposed to the apical surface of peripodial cells, we find it most reasonable to propose that luminal DPP is biologically active. Future studies should attempt to distinguish between the biological action of luminal DPP versus ligand that might be actively transported through the disc epithelium by planar transcytosis.

A Biphasic Model for DPP Action in the Wing Blade Primordium

Previous studies document the action of DPP as a gradient morphogen in patterning columnar cells of the wing

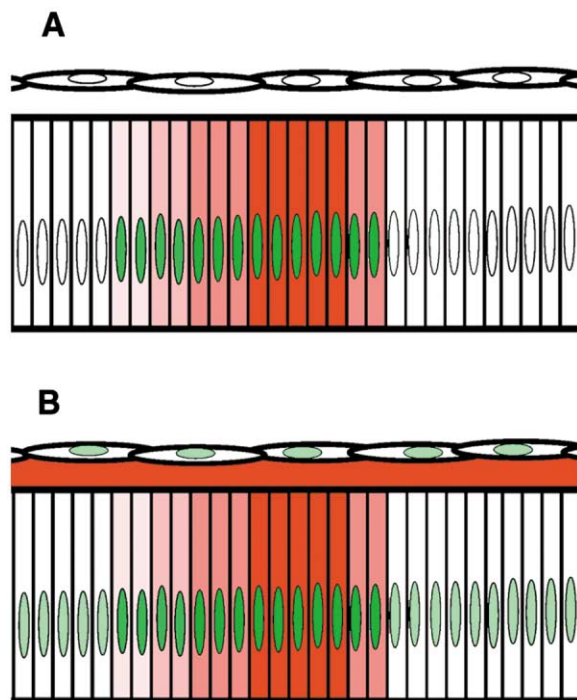


Figure 8. A "Biphasic" Model for DPP Activity in the Wing Blade Primordium

In the cartoon cross-sections (A) and (B), DPP is red and DPP pathway activity is indicated by green nuclei.

(A) In a strictly planar model, DPP is transcribed in a central stripe of source cells at the A/P compartment boundary and secreted laterally into the disc epithelium. High levels of pathway activity are observed centered on the DPP stripe. However, a gradient of DPP in columnar cells does not readily explain the broad requirement for DPP in both peripodial and columnar cell survival.

(B) A biphasic model for DPP action. DPP is secreted laterally into the columnar epithelium and vertically into the lumen. Luminal DPP stimulates sufficient pathway activity to maintain peripodial and columnar cell survival. Just as luminal DPP stimulates low levels of pathway activity in all cells, DPP moving laterally by planar transcytosis generates a high-intensity activity gradient within the columnar epithelium. In columnar cells, the activity profiles stimulated by luminal and epithelial DPP integrate to form the total activity gradient.

blade primordium, although DPP is also broadly required for cell survival (Burke and Basler, 1996a; Moreno et al., 2002). It is not clear how the nearly ubiquitous requirement for DPP signaling within the blade primordium can be reconciled with a steeply graded ligand distribution. Indeed, developing wing discs do not exhibit gradients of cell proliferation or cell death. One plausible explanation for the dual function of DPP in graded patterning and global survival is the existence of spatially discrete ligand pools within developing discs. Our results are consistent with a biphasic model (Figure 8) where high levels of DPP are secreted into the lumen while lower levels form a steep transcytotic morphogen gradient within the columnar cell layer. Functional isolation of the two pools could be achieved if apical junctional complexes act as a diffusion barrier between apical and basolateral membrane domains of the columnar epithelium (e.g., Tepass et al., 2001). According to this model, the luminal fraction of DPP would maintain a low, but

ubiquitous, level of pathway stimulation sufficient to support cell viability throughout wing blade columnar cells and the apposed peripodial epithelium (e.g., Burke and Basler, 1996a; Moreno et al., 2002; Figures 1–3). Simultaneously, the intraepithelial fraction of DPP would activate a steep activity gradient to pattern the central region of the wing blade columnar epithelium in a concentration-dependent manner (e.g., Nellen et al., 1996; Lecuit et al., 1996; Tanimoto et al., 2000; Entchev et al., 2000; Teleman and Cohen, 2000). In the wing blade, the integration of luminal and epithelial stimuli would conform to expectations: a low level of ubiquitous pathway stimulation with a steep activity gradient positioned at the A/P boundary of the disc columnar epithelium.

One counterintuitive feature of this model is that the luminal fraction of DPP appears to be more-highly concentrated than the intraepithelial fraction (Figure 3), yet it stimulates lower levels of pathway activity in columnar cells. This paradox could be explained if DPP pathway stimulation was limited by the surface area of columnar cells exposed to the lumen. Alternatively, differential localization of DPP receptors along a cell's apicobasolateral axis could be critical for determining the amplitude of the DPP response. In general, subcellular restriction of ligand receptors may prove to be a significant factor in understanding tissue-specific responses to secreted signals or pathway specificity in cases where multiple receptors share common signal transduction machinery. Similarly, spatial compartmentalization of extracellular ligands within developing tissues could be an important mechanism for generating diverse cellular responses to the relatively limited number of conserved signaling pathways.

Experimental Procedures

Drosophila Culture and Genetic Crosses

Fly stocks were maintained on standard media at 25°C unless otherwise indicated. For peripodial-specific Gal4 experiments, we employed *c311-Gal4* in the eye (Gibson and Schubiger, 2000) and *AGIR-Gal4* in the wing. On the basis of the analysis with *UAS-nGFP*, both drivers were not expressed in columnar cells during the late second and third larval instars. Because Gal4 driver specificity is a potential concern in such experiments, we also used *teashirt-Gal4* (which is expressed in peripodial cells but repressed in presumptive blade columnar cells) to drive *UAS-dad* and observed similar, if not more extreme, effects on wing disc size (data not shown). Still, we cannot rule out low levels of Gal4 driver expression in columnar cells during embryonic or early larval stages. Similarly, we cannot rule out expression of either driver in columnar cells during pupal morphogenesis.

Clonal Analysis

Standard FRT-mediated somatic recombination was performed as described (Xu and Rubin, 1993) with the mutant alleles *tkv⁵* and *tkv⁷* (Terracol and Lengyel, 1994). Virgin females of the genotype *yw;hsflp¹²²; Ubiquitin-GFP, FRT40A/SM6-TM6B* were crossed to males of the genotype *w; tkv⁷, FRT40A/SM6-TM6B* (Penton et al., 1994) or *w; tkv⁵, FRT40A/SM6-TM6B* (Penton et al., 1994) and heat-shocked for 60 min at 37°C at the times indicated to generate somatic *tkv⁻* clones (both stocks were a kind gift of C. Martin-Castellanos and B. Edgar). Notably, for analysis of clone viability, we analyzed living discs mounted in PBS to improve GFP intensity.

Immunocytochemistry and Imaging

Both live and fixed material were observed with coverslip spacers in order to preserve the integrity of the luminal cavity. Imaginal disc material was fixed, immunostained, and imaged after the methods of

Gibson and Schubiger (1999), except that fixation was 4% paraformaldehyde in PBS. Primary antibodies were used at the following dilutions: guinea pig anti-Discs Large (gift of A. Radovic and P. Bryant), 1:2000; mouse anti-Engrailed 4D9 (DSHB), 1:50; mouse anti-Wingless (DSHB), 1:10; mouse anti-ELAV (DSHB), 1:200; rabbit anti-phosphohistone H3 (Upstate Biotech), 1:2000; rabbit anti-DPP (Panganiban et al., 1990b), 1:100. Secondary antibodies were goat anti-guinea pig Alexa 586 (Molecular Probes), 1:500; goat anti-mouse Bodipy (Molecular Probes), 1:200; goat anti-mouse Texas red (Jackson ImmunoResearch), 1:200; goat anti-rabbit Bodipy (Molecular Probes) 1:200. All rinses were carried out in concave glass culture dishes to reduce the loss of material. Images were collected with either a Leica TCS-NT confocal microscope (Figure 2 and Figures 3A–3D) or a Bio-Rad MRC 600 confocal microscope system. Computer-enhanced 3D reconstructions were assembled from confocal Z-series using the "Import Bio-Rad MRC 600 Z-series" macro in NIH Image 1.62. Figures were assembled using Adobe Photoshop 5.0 and Canvas 5 software.

P{LacZ} to P{Gal4} Conversion

Our transformation-based method for conversion of P{LacZ} strains to P{Gal4} strains was a variation on the genetic logic presented in Sepp and Auld (1999) and the gene targeting approach outlined in Keeler et al. (1996). *w*; *P1615-LacZ* embryos were injected with 500 $\mu\text{g}/\mu\text{l}$ P{w⁺, Gal4} DNA (gift of Ed Giniger) and 100 $\mu\text{g}/\mu\text{l}$ pTurbo. The injection buffer employed was 0.1 mM Na phosphate (pH 7.8) and 5.0 mM KCl in ddH₂O. Red-eyed (*w*⁺) transformant pupae were selected to establish stocks. We obtained 50 viable lines and established a total of 22 distinct eye color stocks from 15 *w*⁺ transformant strains. Transformant stocks were then crossed to virgin *UAS-GFP* flies and progeny were scored for a wing peripodial-specific GFP expression pattern. We identified 2/22 lines in which *Gal4* was expressed in a pattern identical to the original *P{LacZ}* insertion, one of which is *AGIR-Gal4*.

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